

PocketQuake

From a catalog CSV to a relocation notebook in one command

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What is PocketQuake?

A thin orchestrator that takes a list of earthquakes and produces a relocation + focal-mechanism notebook. It glues two specialised submodules:

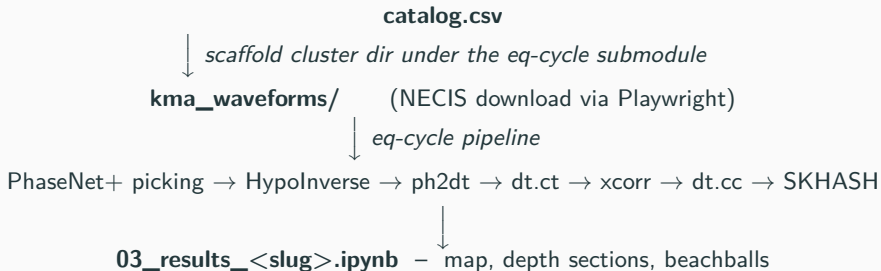
- **seismoseo/necis-downloader** — KMA NECIS waveform fetcher (Playwright + the FTP staging server).
- **seismoseo/korea-cluster-relocation** — the picking + HypoInverse + HypoDD + SKHASH pipeline.

One command, end to end:

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

Output: an executed Jupyter notebook with maps, depth sections, and beachball diagrams for every event you fed in.

Workflow at a glance



You write the catalog and run one command. PocketQuake takes care of dispatching to the right waveform source, registering the cluster, and executing the notebook.

Prerequisites

OS Linux or macOS (bash wrapper; Windows uses WSL)

glibc ≥ 2.28 — Ubuntu 20.04+, RHEL/Rocky 8+, Debian 10+. CentOS 7 / RHEL 7 are below the line (Playwright fails).

Python 3.10+ (pinned by `environment.yml`)

conda Miniforge / Miniconda / Mambaforge

gcc/gfortran ≥ 9.3 (for the compiled externals)

Disk ~ 10 GB (conda env + Chromium + waveforms)

Network HTTPS to `github.com`, `code.usgs.gov`, `conda-forge`, `necis.kma.go.kr`

NECIS account (apply at `https://necis.kma.go.kr`, research / institutional accounts only; approval 1–5 days).

STP account is optional — only for pre-2020 waveforms.

Install [1/2] — Python environment

```
# 1. Clone with submodules
git clone --recurse-submodules https://github.com/seismoseo/PocketQuake.git
cd PocketQuake

# 2. Create the conda env (one command does everything Python-side)
conda env create -f environment.yml
conda activate pocketquake

# 3. Editable installs for PocketQuake itself + the NECIS submodule
pip install -e . -e external/necis-downloader

# 4. Playwright Chromium (one-time, ~200 MB)
playwright install chromium

# 5. Credentials
cp .env.example .env
# then edit NECIS_USER, NECIS_PASS
```

Install [2/2] — External programs

PocketQuake's default chain calls these from the shell. None are pip-installable. Put each binary on \$PATH:

- hyp1.40 — HypoInverse (USGS Fortran)
- ph2dt + hypoDD — Waldhauser's HypoDD (Fortran)
- mseed2sac — IRIS converter (C)
- libarchive / bsdtar — already in environment.yml

Two research clones reachable by env var:

```
git clone https://github.com/AI4EPS/EQNet.git
echo "EQNET_DIR=$PWD/EQNet" >> ~/PocketQuake/.env

git clone https://code.usgs.gov/esc/SKHASH.git
echo "SKHASH_DIR=$PWD/SKHASH/SKHASH" >> ~/PocketQuake/.env
```

Full per-tool source + build instructions: **docs/EXTERNAL_TOOLS.md**. **Pure-Python option** (-python): replaces the Fortran hyp1.40 + ph2dt/hypoDD with HypoSVI + EikoNet and relocDD-py — no Fortran build needed. See the next-but-one slide.

A plain CSV with 10 columns, KST origin times:

```
Year,Month,Day,Hour,Minute,Second,Latitude,Longitude,Magnitude,Depth
2025,2,7,2,35,34,37.14,127.76,3.1,9
2025,2,7,2,54,38,37.14,127.76,1.4,6
2025,2,7,3,49,4,37.14,127.76,1.5,7
2025,2,8,10,13,23,37.14,127.76,1.6,7
```

- Origin time interpreted as KST (UTC + 9 h)
- PocketQuake auto-derives the cluster epicenter (catalog centroid) and the spatial bounds (bbox + 0.2°); override with `-epi`, `-bounds`
- Magnitude is informational — relocation does not use it
- Depth ≥ 0 , measured in km

Run the chungju example

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju
```

The wrapper:

1. scaffolds a cluster dir under
external/korea-cluster-relocation/chungju_cluster/
2. downloads NECIS waveforms (Playwright drives the JS portal, polls the staging queue, streams the FTP files)
3. runs PhaseNet+ → HypoInverse → HypoDD → SKHASH
4. builds + executes the results notebook

Useful flags:

- `-python` — pure-Python relocation backend (no Fortran; next slide)
- `-source mixed` — per-event STP+NECIS dispatch
- `-picker stead` — SeisBench PhaseNet (no EQNet needed)
- `-mainshock <UTC>` — Gwangyang-style mainshock treatment
- `-cores N` — cap xcorr workers (default 10)

High-rate stations: automatic anti-aliasing

PhaseNet+ runs at **100 Hz**, so stations recorded above that (200 Hz KMA/KG, 250 Hz nodal/geophone arrays) are down-sampled first.

- **The trap:** EQNet's reader down-samples by plain *linear interpolation*, *no lowpass* — energy above the new 50 Hz Nyquist *folds* onto the P/S onsets, distorting pick times and first-motion polarity.
- **The fix (v1.12.0):** lowpass *before* the picker — demean → taper → **zero-phase** Butterworth at 45 Hz ($0.9 \times$ new Nyquist) → resample. Zero-phase keeps onsets on time (a causal filter would bias picks late).
- Fires **only when native rate > 100 Hz**: 100 Hz data passes through byte-identical; up-sampling (40 → 100 Hz) is alias-free and untouched.

Verified on a real 200 Hz KG trace: energy above 45 Hz drops 0.064% → 0.003%.

Pure-Python backend (-python)

A Fortran-free relocation chain that drops in for hyp1.40 + hypoDD:

- **HypoSVI + EikoNet** — absolute location by Stein variational inference over a neural-network travel-time field (pretrained weights shipped).
- **relocDD-py** — Katherine Biegel's Python port of hypoDD for the double-difference relocation.

One-time setup (no compiler) — clone the three tools, fetch the weights:

```
git clone https://github.com/katie-biegel/relocDD-py.git
git clone https://github.com/Ulvetanna/HypoSVI.git
git clone https://github.com/Ulvetanna/EikoNet.git
cat >> ~/PocketQuake/.env <<ENV
RELOCDD_PY_DIR=$PWD/relocDD-py
HYPOSVI_DIR=$PWD/HypoSVI
EIKONET_DIR=$PWD/EikoNet
ENV
python -m pipeline.core.fetch_eikonet    # pretrained kim1983 + kim2011
```

```
./pocketquake.sh ./examples/chungju/chungju_catalog.csv chungju --python
```

What you see while it runs

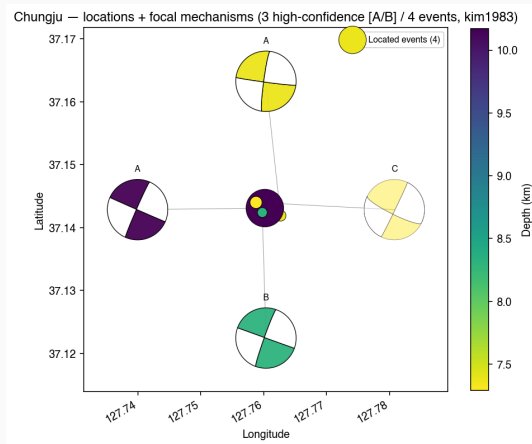
Foreground (-fg): full log to terminal.

Default (background): nohup.out in the working directory.

```
[pocketquake] scaffolding chungju_cluster ...  
[necis] login OK; submitting request for 4 event(s)  
[necis] request id 4f3a... status=P -> C, downloading 4 SAC bundles  
[eqcycle] picking with phasenet_plus (~30s/event)  
[eqcycle] hypoinverse: 4 events located  
[eqcycle] xcorr 6 pairs (10 workers) ...  
[eqcycle] dt.cc -> hypoDD reloc OK (4 events)  
[eqcycle] focal_mechanism: 4 SKHASH solutions  
[pocketquake] executing 03_results_chungju.ipynb
```

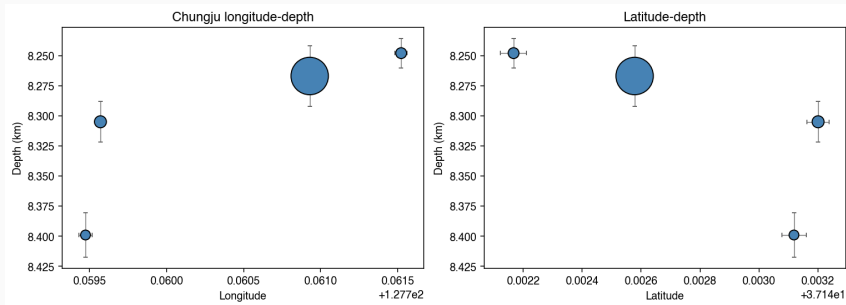
Wall-clock for chungju: ~15 min on a 16-core box.

Output 1 — Relocated map + focal mechanisms



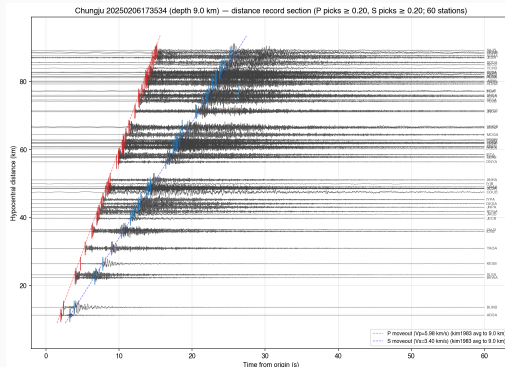
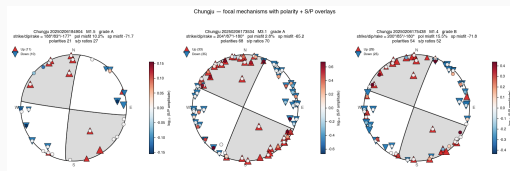
Four located events (depth-coloured dots) at (37.142°N , 127.760°E) with quality A/B/C SKHASH beachballs drawn on a leader-line ring. All four solutions converge on a near-vertical N–S right-lateral strike-slip plane.

Output 2 — Depth sections



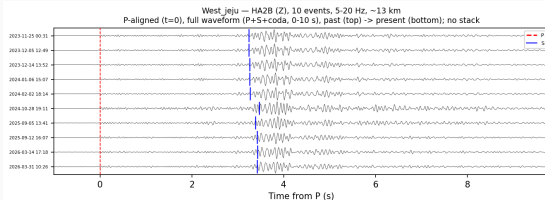
Three orthogonal cross-sections (E–W, N–S, along-strike) showing the depth distribution. HypoDD relative-relocation tightens the cluster from ~ 1 km scatter (KMA bulletin) to ~ 200 m.

Output 3 — Beachball gallery + record section

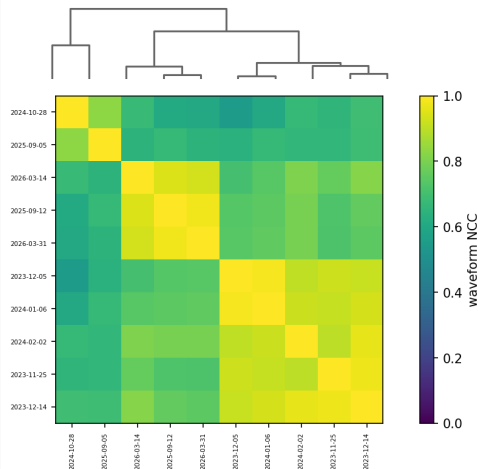


The notebook bundles a per-event beachball gallery (left) and a record-section view of the M3.1 mainshock with PhaseNet+ pick markers, first-motion polarities, and the SKHASH-inferred S/P amplitude ratios (right).

Output 4 — Waveform similarity (per dt.cc sub-cluster)



West jeju — HA2B (Z) waveform CC matrix
10 events, 5-20 Hz, full waveform (hierarchical clustering order)



Run on your own catalog

1. Save your KMA-style CSV (the format from slide 6).
2. One command:

```
./pocketquake.sh ~/my_catalog.csv myswarm
```

3. Override the auto-derived epicenter / bounds if your catalog spans a wide region:

```
./pocketquake.sh ~/my_catalog.csv myswarm \  
  --epi 35.46,128.43 \  
  --bounds 35.3,35.65,128.25,128.65
```

4. Mainshock treatment (re-runs xcorr→dt.cc with the mainshock excluded, builds a _main notebook):

```
./pocketquake.sh ~/my_catalog.csv myswarm \  
  --mainshock 20240912144719
```

Command-line options (`./pocketquake.sh --help`)

```
--python pure-Python backend (HypoSVI + relocDD-py; no Fortran)
--compare run Fortran and Python on the same picks → side-by-side notebook
--source necis | stp | mixed waveform source (default necis; use stp/mixed for pre-2020 events)
--skip-download reuse waveforms already on disk (skip the slow download)
--skip-pipeline scaffold + download only (no relocation / notebook)
--mainshock <UTC> add Gwangyang-style mainshock treatment (_main notebook)
--picker phasenet_plus | stead picker model (stead needs no EQNet)
--velmodel kim1983 | kim2011 velocity model for relocation + focal mech + notebook
--cores N cap xcorr workers (default ~10)
--epi, --bounds override the auto-derived epicenter / region
--fg foreground (default: background, logs to <slug>_run.log)
```

Velocity model: location always computes both kim1983 + kim2011; `--velmodel` (default kim1983) selects which drives the relocation + focal mechanisms + notebook. Fully for the Fortran path; with `--python`, HypoSVI keeps its bundled EikoNet weights.

Common pitfalls (docs/TROUBLESHOOTING.md)

python interpreter not found set POCKETQUAKE_PYTHON=/path/to/python in .env or put python3 on \$PATH.

Permission denied (publickey) use the HTTPS clone URL, not SSH.

Submodule path '...' not registered you forgot -recurse-submodules; fix with git submodule update -init -recursive.

EQNET_DIR not set PhaseNet+ is the default picker; either install EQNet (slide 5) or pass -picker stead.

libstdc++ version not found glibc/libstdc++ too old (CentOS 7); upgrade OS or run the NECIS download elsewhere.

NECIS_USER empty fill in .env.

Where to go next

- **Read the executed result notebook** —
`external/korea-cluster-relocation/pipeline/notebooks/03_results_<slug>.ipynb`
- **Inspect the auto-derived configuration** —
`external/korea-cluster-relocation/pipeline/clusters/<slug>.py`
- **Tune the run** — `./pocketquake.sh -help`
- **Other examples bundled** — `examples/sangju/`, `examples/chungju/`, the cluster modules registered in `korea-cluster-relocation/pipeline/clusters/`
- **Docs** — `docs/INSTALL.md`, `docs/EXTERNAL_TOOLS.md`, `docs/TROUBLESHOOTING.md`

<https://github.com/seismoseo/PocketQuake>

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